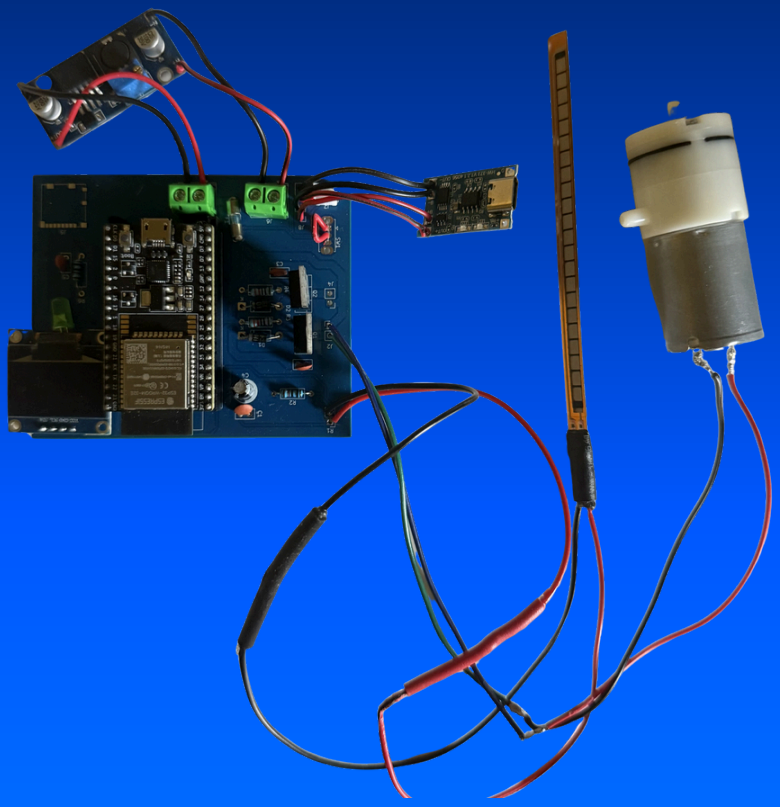
MEDICAL INNOVATION

PNEUFLEX



PRAYUSH BHATTARAI
JISHNU SAI KONDA
EVAN LOI
MALHAR KURLE

SECONDARY DIVISION
#26209
CHEROKEE TRAIL HIGHSCHOOL
COLORADO COMMUNITY COLLEGE SYSTEM
COLORADO HOSA

The Problem Space

THE CLINICAL REALITY

Muscle Atrophy

Gradual **loss** of muscle tissue from disuse, aging and neurological conditions such as stroke, spinal cord injury, ALS or muscular dystrophy. Atrophy **weakens** muscles, limits mobility and increases **dependence** on caregivers.

Effects of Atrophy

Upper limb atrophy **impairs** eating, dressing, hygiene and writing, all basic daily necessities. A **decline** in strength leads to social withdrawal, reduced independence and can often significantly **lower** quality of life.

Treatment Limitations

Current **methods** rely on physical therapy, medications and assistive devices. Progress is **slow**, **expensive** and only **partially** effective. This ends up leaving patients **underserved** with lasting mobility challenges.

INNOVATION CONCEPT: PNEUFLEX

PneuFlex is a **low-cost,** wearable **pneumatic (1) rehabilitation sleeve** designed to assist upper arm movement for individuals who suffer from **muscle atrophy** or limited mobility. The system uses a soft **McKibben actuator (2)**, embedded **flex sensors**, an **ESP32E** microcontroller and adaptive control software within a lightweight sleeve. PneuFlex includes real-time monitoring and data logging capabilities with a **companion app** and **TinyML AI** adaptive software to support rehabilitation tracking. PneuFlex was designed for accessibility and scalability, providing affordable assistive support for **home therapy** and clinical environments while also promoting improved functional independence.

OUR MOTIVATION

We created PneuFlex because accessible rehabilitation solutions are limited. A close friend of ours, **Forrest**, suffered an injury that directly led to **muscle atrophy**. His muscles deteriorated from inactivity, making movement **strenuous**. Throughout recovery, the lack of consistent, accessible rehabilitation treatments directly contributed to a **lengthy recovery** process. This was our inspiration for creating PneuFlex.



THE REHABILITATION GAP

Motor recovery after stroke or injury depends on how many **repetitions** are done per hour for recovery. However, the current care model makes it **difficult** to achieve the 300-400 reps/hr required:

REPS NEEDED/HR	300-400 reps/hr
UNAIDED	10-20 reps/hr
THERAPY	80-150 reps/hr
PNEUFLEX	250-350 reps/hr

COST BARRIER

Clinical sessions cost $150–$350 each, severely **limiting** how often patients can attend. Most receive therapy only 2–3× per week which is **below** the repetition density **(3)** required for meaningful neuroplastic recovery **(4)**.

EXISTING SOLUTIONS & FAILURES

Passive Braces
No **active** assistance. No AI. Provides support but does not drive neuroplastic recovery. Limited **efficacy**.

Electrical Stimulation (FES) (5)
Painful, requires clinical setup, **inconsistent** across patients with significant denervation **(6)**. Does not assist **voluntary** movement.

Robotic Exoskeletons (7)
$50,000–$100,000. Rigid, **clinic-confined**, requires trained operators. **Zero** viable home-use pathway for patients.

PneuFlex (Target Solution)
Active, AI-personalized, pneumatic home therapy at $90. Battery-powered. Compliant-safe. Wearable **anywhere**.

WHO IS AFFECTED/TARGET MARKET

Stroke Survivors
795K strokes/yr in US. Upper limb weakness affects **77%** of stroke survivors, the #1 driver of **long-term** disability.

Spinal Cord Injury
SCI patients with motor signals **benefit** from assisted repetition to **preserve** and **strengthen** neural pathways.

ALS/Muscular Dystrophy
Conditions where assistive movement maintains functional capability and **slows atrophy** progression during early stages.

Post-Surgical Recovery
Patients **recovering** from surgery requiring controlled, low-resistance repetitive motion to **rebuild** strength.

FEATURE COMPARISON MATRIX

FEATURE	Brace	FES	Exo	PneuFlex
Active assist	✗	✓	✓	✓
AI intent detection	✗	✗	✗	✓
Home use	✓	✗	✗	✓
Under $500	✓	✗	✗	✓
Data logging	✗	✗	✗	✓
Mechanically safe	✓	✓	✗	✓

System Architecture

SIGNAL TO ACTUATION PIPELINE

01	02	03	04	05
Flex Sensing	**Windowed Input**	**AI Inference**	**Actuation**	**Log & Sync**
▶️ Flex resistor on the elbow joint **changes** resistance as the **arm bends**. Sampled at 100Hz via voltage divider into ESP32 ADC.	▶️ 500ms rolling window of sensor data fed to TinyML model. Spectral features are **extracted** from time-series **signal**.	▶️ Edge Impulse model classifies: **Intentional** Flex, Extension, or Idle. Threshold: **≥85%** confidence.	▶️ MOSFET fires a 12V pump which **inflates** the **McKibben**, **assisting** elbow flexion. When extended: solenoid vents air.	▶️ Session data **logged** to SD card. Synced via WiFi to cloud. OTA **(8)** firmware **updates** via **companion app**.

HARDWARE COMPONENTS

MICROCONTROLLER	ESP32-DevKitC-32E — dual-core, FreeRTOS, WiFi + BLE
SENSOR	Resistive flex sensor @ 100Hz sampling via ADC
ACTUATOR	McKibben pneumatic artificial muscle (braided mesh + bladder)
SWITCHES	2× IRLZ44N logic-level MOSFETs (3.3V gate drive)
PROTECTION	Flyback diodes, 2A fast-blow fuse, gate resistors
POWER	3.7V 2200mAh LiPo, XL6009 boost 12V; TP4056 charger
DISPLAY	0.96″ SSD1306 OLED — live stats: angle, confidence, reps
STORAGE	16 GB micro SD card — local session logging
PCB	Custom 2-layer KiCad PCB — manufactured by JLCPCB

COMPANION APP

The **PneuFlex** companion app connects via **Bluetooth Low Energy (BLE)** and serves as the bridge between the patient, therapist, and device. Once a session is complete, data syncs to the cloud over **WiFi** for longitudinal tracking **(9)**. This enables evidence-based clinical decision-making without requiring a clinical visit.

Patient Display — Large-font **rep counter**, movement quality score (0-100%), session timer, and weekly progress chart. Designed for **accessibility** so that no technical knowledge is required. Displays trend vs. the previous week to motivate **progression**.

Therapist Display — **Real-time** joint display, activation confidence, actuation timing log, rep quality, and full CSV export per session. The therapist can **adjust** parameters (rep target, session duration, confidence threshold, etc) **remotely** without the patient returning to the clinic.

Security — **PIN-locked** therapist configuration panel, Cloud firmware updates, Cloud sync toggle, confidence threshold adjustment, and **local storage** of data on the phone.

Bluetooth
Live data output (rep count, confidence %, joint angle, pump state) synced to app in real time.

Wifi · Cloud
Post-session upload of full session log to the cloud, visible longitudinal recovery trends, and firmware update.

MCKIBBEN ACTUATOR PHYSICS

A McKibben actuator is a device that converts **air pressure** into **movement**. A braided mesh sleeve surrounds an elastic inner tube. Pressurization causes radial **expansion** constrained by the mesh, forcing the actuator to shorten longitudinally, generating a **pulling force** similar to biological muscle contraction. Positioned along the inner arm, it assists elbow flexion while remaining compliant. Resistance is **limited** to a certain extent to maintain safety.



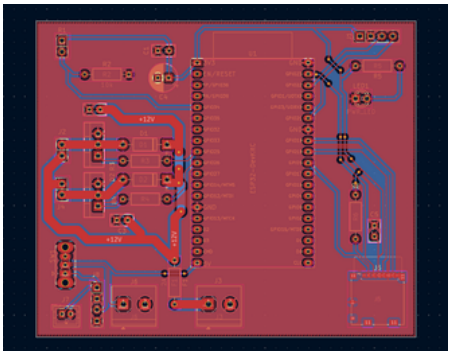
APP INTERFACE



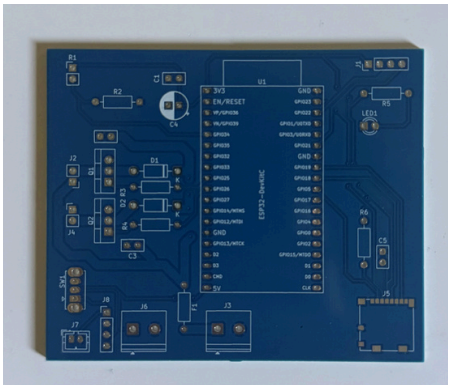
- **Rep Counter**
  Live, large-font count & each assisted rep is logged instantly over BLE
- **Quality Score Ring**
  A composite metric from AI condifence % +movement smoothness. Ring fills as quality improves
- **Session Timer**
  Counts session time. Therapist sets duration limit remoely via *Settings Mode*
- **Weekly Progress Bar**
  Cumulative reps vs. weekly target. Tracks weekly total rep progress.

Physical Device & Build Roadmap

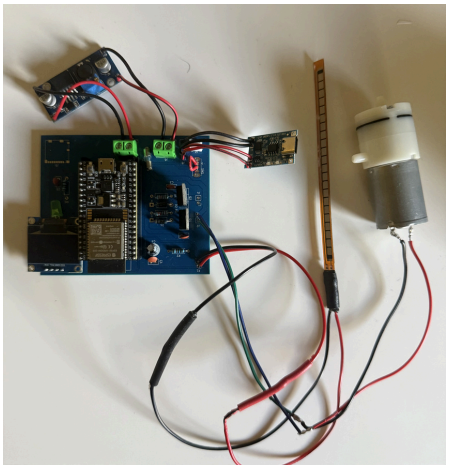
PCB Design



PCB Fabricated

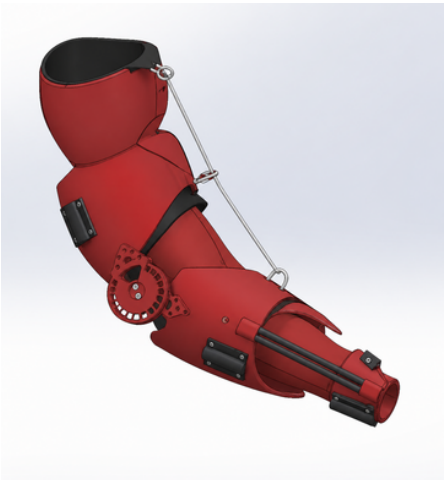


PCB Assembled



CAD Model

(Not to scale & adjusted after printing)



DEVELOPMENT ROADMAP

COMPLETE ✓
Circuit Design & PCB Layout Full schematic in KiCad. 2-layer PCB designed with MOSFET switching, boost converter, TP4056, OLED, SD card, and ESP32 breakout footprints. Gerbers exported.

COMPLETE ✓
PCB Fabrication PCB was sent to be manufactured by JLCPCB. Once the boards were received, components were populated.

COMPLETE ✓
PCB Assembly Assembly Core functionality: flex sensing, pump control, battery powered, McKibben actuator, flex threshold and limit hard coded.

COMPLETE ✓
Sleeve Printed Integrates OLED display, Bluetooth, WiFi, TinyML **(10)** model trained on Edge Impluse, boost converter circuit, ESP32.

COMPLETE ✓
Clinical Validity Controlled trials measuring rep count, fatigue reduction, and independence scores across a 4-week testing period.

V1 FEATURE SET

- ✓ Flex Sensing (100 Hz)
- ✓ Pump Control
- ✓ Controlled Actuation
- ✓ Battery Powered

POWER BUDGET ANALYSIS

COMPONENT	CURRENT DUTY
ESP32 (ACTIVE)	240mA 100%
12V PUMP (ACTUATION)	800mA ~15%
OLED DISPLAY	20mA 100%
EST. SESSION LIFE	~3-4 hrs 2200mAH

V2 PCB IMPROVEMENTS

OLED Live Display
Shows reps, joint angle, and confidence on the sleeve.

Bluetooth LE (BLE)
Wirelessly streams live session data to the companion app.

WiFi Cloud Sync
Uploads session logs to the cloud after each session.

TinyML Inference <2ms
Classifies movement intent in under 2 milliseconds on-device.

9-Layer Safety System
Nine safeguards preventing overcurrent, over-inflation, and unintended actuation.

SD Card Data Transfer
Micro SD footprint directly on PCB to remove breakout module.

TinyML & Safety

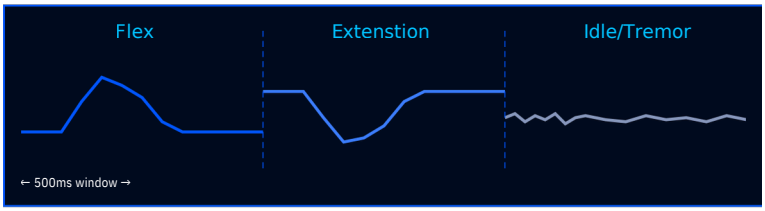
WHY USE AI?

Stroke and neurological patients exhibit **highly variable** sensor readings. Many have different tremor signatures, movement speeds, and flex values. A fixed threshold to activate the device would be very inconsistent. The TinyML model is **personalized per-user** via a calibration session, learning each individual's specific intentional “flex signature”.

MODEL PIPELINE

INPUT	500ms rolling window of raw flex sensor time-series data sampled at 100Hz	50 samples per window
FEATURES	Spectral analysis (FFT): Frequency domain features distinguish intentional motion from tremor/noise	DSP Features
MODEL	Edge Impulse trained classifier quantized **(11)** to 8-bit integers for embedded execution	INT8 Quantized
OUTPUT	3-class classification: Flex , extension or Idle. Acts only when classification is above 85% confidence	≥85% threshold
LATENCY	Full inference runs on ESP32 in under 2ms. No cloud means no connectivity dependency	<2ms / cycle
FOOTPRINT	500ms rolling window of raw flex sensor time-series data sampled at 100Hz	4-6 KB Flash

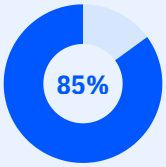
SIGNAL CLASS VISUALIZATION



EDGE CASE HANDLING

Parkinson Tremor	High-freq spectral signature filtered by fast fourier transform **(12)** features. This is classified as Idle
Accidental Contact	Short transient, lacks 500ms intentional ramp — below confidence gate
Fatigue Drift	Recalibration prompt appears after 3 consecutive low- confidence reads

PERFORMANCE



False Positive Reduction

TinyML reduced **unwanted** actuations by ~85% compared to simple threshold triggering which is critical for **patient safety** and trust in the device during rehabilitation.

Personalization via Calibration

At first use, the patient performs guided flex/extension cycles. Labeled sensor data is captured, and the Edge Impulse model **retrains** to the individual's **unique** movement signature. This handles: tremor filtering, baseline offset variation, and movement speed differences between patients.

9 LAYER SAFETY ARCHITECTURE

H — 2A Fast-Blow Fuse
Hardware overcurrent protection on 12V rail

H — Flyback Diodes
Protect MOSFETs from inductive voltage spikes

H — Gate Resistor
Limit switching transients, reduce EMI

H — TP4056 Protection
Battery overcharge and short-circuit guard

F — Watchdog Timer
Firmware restarts on any system hang

F — Max Actuation Limit
Hard-coded duration cap that prevents over-inflation

F — Min Interval Lock
Enforces rest between actuations + prevents injury

F — Emergency Stop
Physical button immediately vents actuator

M — Mechanical Compliance
McKibben actuator yields under resistance. This means that it cannot force a joint beyond its range of motion

SYSTEM PERFORMANCE

85% FALSE POSITIVE REDUCTION	**9** DIFFERENT SAFETY LAYERS
<2ms LATENCY	**4 hr** BATTERY LIFE

1: Powered by compressed air rather than motors or electricity
2: A soft artificial muscle that shortens and pulls when inflated with air
3: The number of assisted movement cycles per hour
4: The brain's ability to rewire itself after injury through repeated movement
5: A therapy that uses electrical current to trigger muscle contractions
6: The loss of nerve supply to a muscle, making electrical stimulation ineffective
7: A rigid, motorized wearable frame that assists limb movement.
8: Software updates delivered remotely via WiFi without a clinic visit
9: Monitoring a patient's data over time to observe recovery trends
10: An AI that runs locally on low-power microcontrollers
11: Compressing an AI model's precision to runs on low-power hardware
12: A mathematical process that separates movement signals from tremors

Transforming Rehabilitation

795K
STROKES / YEAR (US)
Annual **US incidents** relating to stroke. Each patient is a **potential PneuFlex** candidate

$34B
STROKE REHAB COSTS
Annual US stroke rehabilitation **spending** (dominated by in-person session costs)

25%
VISIT REDUCTION TARGET
Estimated clinical visit **reduction** when **PneuFlex** enables sufficient **home** supplemental therapy

<1 WK
RECOUP COSTS
PneuFlex recovers its $90 **cost** within the **first week** of usage through avoided clinical sessions

SYSTEMIC HEALTHCARE SHIFT

PneuFlex reveals a fundamental shift in rehabilitation delivery from episodic clinical visits to **continuous** home-based therapy. By **increasing** the frequency and consistency of movement, it helps reduce disuse muscle atrophy while supporting sustained **neuromuscular engagement**. This approach enhances neuroplastic recovery by allowing patients to perform high-repetition therapy aligned with their natural recovery pace, rather than being **limited** by the frequency of clinical visits.

NEUROPLASTICITY

Neuroplasticity is the brain's ability to **reorganize** itself by forming new neural connections, allowing it to learn, adapt, and recover from injury. The recovery rate **directly scales** with assisted-repetition density. **PneuFlex** allows for 200+ reps/hour, daily, from home, at no additional cost.

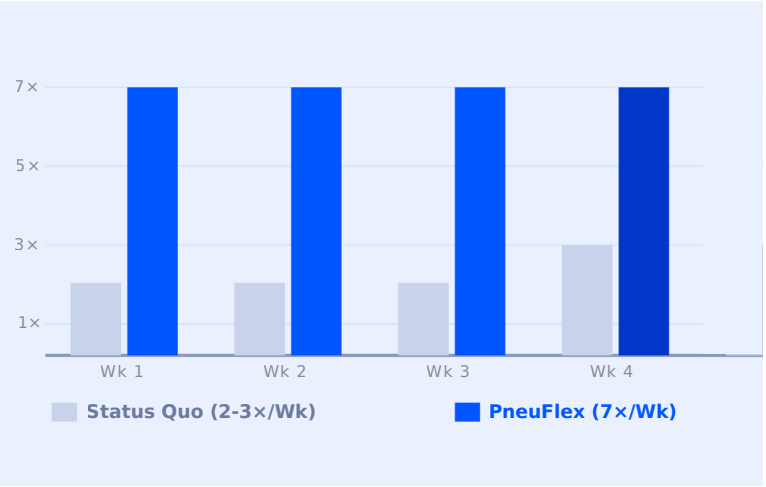
EXPANSION OF REHAB ACCESS

- Rural patients gain **access** to daily rehabilitation without traveling to clinics which **eliminates** geography as a healthcare barrier
- Low-income patients who are **uninsured** for frequent therapy sessions gain therapeutic equality through a **one-time $90 device**
- Therapists gain longitudinal data streams, enabling evidence-based session **customization** rather than **periodic** snapshot assessments
- Healthcare systems reduce total rehabilitation episode cost by **replacing** expensive clinical sessions with AI-guided, **home therapy**

EVIDENCE: REP DENSITY & OUTCOMES

KEY FINDINGS	
Bütefisch et al. (1995)	Repetitive training **improves** recovery in the area affected by paralysis, with **greater intensity** leading to better outcomes
Lum et al. (2002)	Robotic-assisted repetitive movement is **superior** to traditional therapy for upper-limb recovery after a stroke
Winters et al. (2015)	A proportional recovery model **(1)** is developed in which the intensity of assisted practice is able to **predict** the magnitude of recovery in half a year

SESSION FREQUENCY & REP VOLUME



7×/Wk
SESSION FREQUENCY

10×
MORE REPS PER WEEK

TARGET MARKETS

PRIMARY - PATIENTS WITH:
- Post-stroke upper limb weakness
- Partial motor control
- Neuroplasticity window open

SECONDARY - PATIENTS WITH:
- Incomplete spinal injury **(2)**
- Post-surgical elbow rehabilitation
- Early stage ALS

TRAINING REQUIREMENTS

Initial Onboarding (~20 minutes)

Occupational therapist guides the patient through: sleeve wearing, calibration session, app pairing over BLE, mode selection, and emergency stop demonstration. **PneuFlex** requires **0 technical** background from the patient.

Patient Training

App guided calibration at session start. After the initial onboarding, there is **no required** clinical supervision.

Therapist Training

Remote parameter configuration, session data export, and longitudinal trend review. **No hardware training** needed.

Healthcare Careers Transformed

PNEUFLEX AND HEALTHCARE OCCUPATIONS

Occupational Therapy
Occupational therapists **define** therapy standards and interpret session data to **optimize home rehabilitation**

Physical Therapy
PneuFlex enables physical therapists to **enhance** exercise programs, track neural recovery, and adjust clinical visit frequency

Neurology
Neurologists are able to leverage continuous longitudinal sensor data to **assess** motor recovery progression and **guide treatment**

Biomedical Engineering
Engineers develop **integrated systems** by designing advanced actuators, improving sensing accuracy, and developing AI processing

Digital Health PMs
Product managers drive **companion app development**, patient engagement UX, and clinical workflow integration across systems

FDA Regulatory Affairs
Regulatory specialists **handle** clearance, classification, and compliance for wearable medical devices such as **PneuFlex**

PRACTICIONER TRAINING REQUIREMENTS

Patient
20-minute on-board session with occupational therapist. App-guided daily calibration after the session, and no technical skill is required.

Therapist
3-hour training module that covers device prescription, data interpretation, remote configuration, and safety protocols.

Clinician
Review session reports and **analyze** longitudinal outcome trends that are integrated into existing EMR **(3)**

FUTURE PATHWAY

PneuFlex is a **foundation**, not a ceiling. The architecture can be extensible and **transform** engineering and design roles in healthcare.

Whole-Arm Rehabilitation — extend to shoulder, wrist, and finger joints with modular actuator units

Lower Limb Adaptation — McKibben actuators applied to the knee/ankle for gait rehabilitation post-stroke

EMG Integration — replace the flex sensor with surface electromyography **(4)** to allow for earlier detection

Custom ASIC (5) Design — purpose-built silicon replacing ESP32 to reduce power and board area to form a production-grade device

REGULATORY PATHWAY

PneuFlex would likely qualify as a Class II **(6)** medical device under **FDA jurisdiction**, pursuing 510(k) clearance **(7)** by demonstrating substantial equivalence to predicate **(8)** wearable rehabilitation devices. CE marking **(9)** in the EU would follow similar technical documentation **requirements**.

DATA FLYWHEEL EFFECT

Each **PneuFlex** device in the field generate session data. Aggregated at scale, this creates a powerful dataset for:

1. Population-level motor recovery benchmarking across diagnoses
2. AI improvement using federated learning **(10)** from thousands of users
3. Outcomes research — peer-reviewed evidence for clinical validation
4. Predictive modeling — early identification of patients at risk of plateau

GLOBAL HEALTH EQUALITY IMPACT

RURAL US
Patients **60+ miles** away from a **rehab clinic** gain daily therapy access. Geography stops being a **death sentence** for recovery outcomes.

LOW-INCOME
Uninsured patients who cannot afford **$250/therapy** session are able to receive daily rehabilitation for a one-time purchase of **$90**.

LMICs
Low-and-middle-income countries with minimal rehab infrastructure gain access since exoskeletons are **non-existent options**.

ELDERLY
Patients **unable** to travel frequently can use **PneuFlex** at home with remote therapist oversight via app, reducing the **burden** of transportation.

GLOBAL BURDEN CRISIS

15M	85%	>$100
STROKES/YEAR WORLDWIDE	OF CASES OCCUR IN LMICS	FOR ANY TREATMENT OPTION

1: Predicts that recovery magnitude scales directly with therapy intensity.
2: Partial damage to the spinal cord where some signals still pass through
3: A digital version of a patient's chart used within a healthcare system
4: Measures electrical activity produced by muscles
5: Microchip custom-designed for a specific use rather than general computing
6: An FDA category for moderately complex devices requiring special controls
7: FDA approval based on proving equivalence to an already-approved device
8: An already FDA-approved device that a new device claims equivalence to
9: Certification required to legally sell medical devices in Europe
10: AI training across many devices without raw data ever leaving those devices

Restoring Independence

DAILY ACTIVITIES

01

Eating Independently

Elbow flexion is fundamental to self-feeding. **PneuFlex** assists the lift of utensils and cups, restoring a core dignity-preserving daily activity without dependence on a caregiver.

02

Writing & Communication

Holding a pen or a phone requires sustained control of the elbow. **PneuFlex** supports simple motor tasks, allowing patients to write, text, and communicate with others.

03

Personal Hygiene

Brushing teeth, washing the face, and combing hair are all tasks that require repeated elbow flexion. Active assistance helps patients perform independent hygiene, reducing caregiver burden.

04

Dressing & Grooming

Getting dressed independently is a fundamental marker of autonomy. Assisted elbow movement enables patients to pull on shirts, button clothing, and style their appearance.

IMPACT OF RECOVERY

Functional recovery is inseparable from mental health. As muscle atrophy worsens, patients experience a cascade of psychological consequences that **PneuFlex** directly addresses:

LOSS → RECOVERY	Social withdrawal from inability to participate in activities transforms into re-engagement as functional capability returns
RELIANCE → INDEPENDENCE	Caregiver reliance for daily activities develops into restored self-sufficiency which reduces depression and improves self-efficacy **(1)**
SLOW → ACCELERATED	Frustratingly slow plateau-prone **(2)** recovery changes into measurable progress tracked daily, motivating continued engagement

CAREGIVER IMPACT

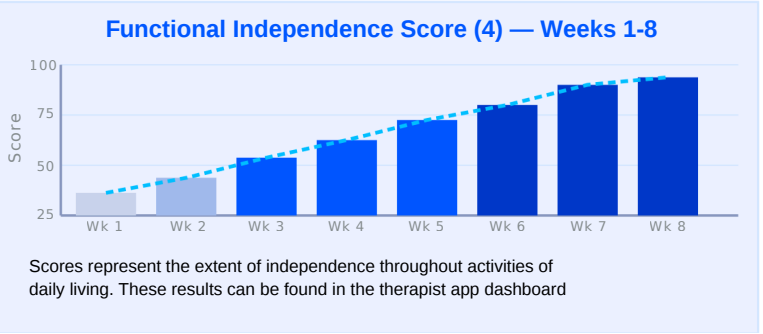
Informal caregivers for stroke survivors experience significantly elevated rates of burnout, depression, and health decline. **PneuFlex's** goal of restoring patient independence directly reduces strain on caregivers, improving the quality of life for entire family systems, not just the patient.

CAREGIVER BURDEN REDUCTION

40%
ESTIMATED ADL **(3)** INDEPENDENCE GAIN

3.5×
MORE REHAB SESSIONS PER WEEK

WEEKLY RECOVERY CHART



Scores represent the extent of independence throughout activities of daily living. These results can be found in the therapist app dashboard

PATIENT DOMAINS RESTORED

Physical Autonomy

Ability to perform ADLs (Activities of Daily Living) without caregiver assistance is essential to autonomy.

Social Reintegration

Return to social activities, work, and family life. This is correlated with improved depression and anxiety scores.

Cognitive Engagement

Goal-directed therapy with measurable progress trackers. This helps combat post-stroke depression.

Family System Health

Reduced caregiver burden protects family mental health. This leads to improved wellbeing for the entire household.

PATIENT PROGRESS INTERFACE



Patient Journey

A DAY IN THE LIFE

7:00 AM Patient puts on **PneuFlex** sleeve. App prompts **2-minute calibration** session. Device learns today's resting muscle state.

7:30 AM Morning therapy session is **~30 minutes** with **~200 assisted reps**. OLED shows live rep count and movement quality score.

12:00 PM Patient eats lunch and complete normal activities with **PneuFlex** assisting as needed, activating only on **intent detection**.

3:00 PM 20 minute **functional** task practice while therapist reviews session **analytics** remotely via app dashboard.

Evening **Session data syncs** to cloud over WiFi and weekly progress chart updates. The patient is able to track progress through app.

INDEPENDENCE MILESTONES

Week 1–2: Feeding
Assisted eating in order to reducing mealtime dependence

Week 3–4: Hygiene
Independent hygiene practices is a milestone for dignity

Week 5–6: Communication
Restoration of ability to maintain communication with others

Week 7+: Home Independence
Reduced clinical visits and restored functional independence

PATIENT FEEDBACK IMPLEMENTATION

OLED	Live rep count and AI confidence to provide immediate feedback per rep
APP	Weekly progress chart and quality score trend to motivate adherence
HAPTIC	Actuation provides feedback to help strengthen motor-sensory loop **(5)**
LED	Green LED on the board pulses when a nerve-detection occurs

3D PRINTED SLEEVE

Custom CAD Polylactic Acid (PLA) Sleeve

The sleeve is custom-modeled using CAD and FDM **(6)**. It is then 3D printed in PLA **(7)**. The parametric model can be printed in different arm sizes, enabling a personalized fit for any patient without the cost of custom production. The sleeve weighs ~300g. PLA is rigid enough to hold actuator alignment while remaining lightweight and skin-safe for extended wear.

RESTORED CONFIDENCE

BEFORE	AFTER
• Dependence	• Independence
• Frustration	• Improved motivation
• Social withdrawl	• Re-engagment
• Increased anxiety	• Reduced burden

THERAPY PROTOCOL

Session Duration	20–45 min, adjustable in therapist app settings
Rep Target	200–400 assisted reps/session
Rest Interval	2s enforced rest between actuations
Max Pressure	Hard-capped actuation duration prevents over-inflation
Confidence Gate **(8)**	≥85% confidence required to trigger actuation

MEASURED OUTCOME

200+	7×	<2ms
REPS/SESSION	SESSIONS/WEEK	LATENCY

12 WEEK STRATEGY

Wk 1–2 Supervised onboarding with OT present for first 2 sessions

Wk 3–6 Independent home sessions and weekly therapist check-in

Wk 7+ Gradual support reduction and monitor outcomes

1: A person's belief in their own ability to accomplish tasks
2: Tending to stall in progress rather than improving steadily
3: Routine self-care tasks like eating, dressing, and bathing
4: Metric tracking how well a patient performs daily tasks without assistance
5: Feedback between the brain's commands and the body's physical sensations
6: 3D printing method where melted filament is deposited layer by layer
7: A lightweight, rigid, skin-safe bioplastic used to 3D print **PneuFlex**
8: A threshold requiring the AI to be at least 85% confident before any actuation

The Cost Revolution

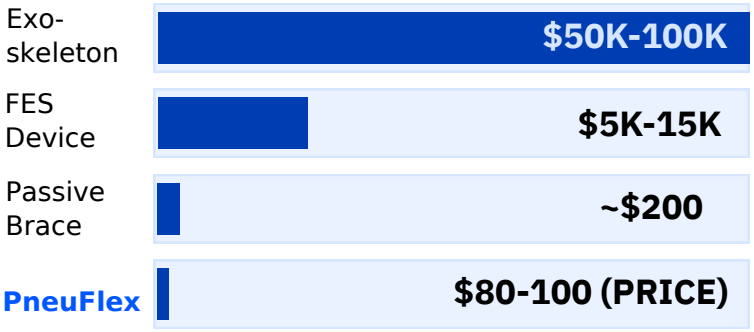
$67 Cost of Materials

$90 Price of PneuFlex

$100K Cost of Exoskeleton

99.8% Percent Cheaper

COST COMPARISION



MATERIAL COST BREAKDOWN

Part	# Cost
ESP32-DevKitC-32E	1× · $8
Custom PCB (JLCPCB, 5-pack)	1× · $.50
McKibben Actuator (DIY)	1× · $.30
12V Diaphragm Pump	1× · $10
Solenoid Valve	1× · $10
LiPo 3.7V 2200mAh + TP4056	1× · $12
XL6009 Boost Converter	1× · $1
IRLZ44N MOSFETs (×2)	2× · $.70
SSD1306 OLED + Flex sensor	1× · $16
Passives, Wiring, Sleeve, Misc.	$8
Total Cost:	$67.20

COST ACCESSIBILITY

Uninsured US	**$90** one-time vs. **$250/session**. Breaks even after first avoided clinical session
Medicare	**Post-510(k)** coverage: **$0** out-of-pocket for qualifying patients
LMICs	**$90** represents price point in markets where exoskeletons are unavailable

ROI ANALYSIS

HEALTHCARE SYSTEM ROI (1)

Average clinical therapy session: **$250**

PneuFlex home session cost: **$0** (device already purchased)

Break-even **(2)**: **device pays for itself** within 24 hours of use

PneuFlex reduces visits by ~25% over 12-weeks: **patient saves $750+**

Net ROI per patient: **~7.7× device cost**

INSURANCE & REIMBURSEMENT

Upon **FDA 510(k)** clearance, **PneuFlex** could qualify for reimbursement **(3)** under **DME** (Durable Medical Equipment) codes — making it effectively **$0** out-of-pocket for most insured patients. This eliminates the financial barrier entirely and positions **PneuFlex** as a standard-of-care prescription device.

VOLUME MANUFACTURING (4)

Prototype (1 Unit)	$67
Small batch (100 units)	~$55/unit
Volume (10,000+ units)	~$25–45/unit

MANUFACTURING COST DRIVERS

PCB Fabrication — JLCPCB
A pack of 2-layer boards costs **~$2.50** total via **JLCPCB**. At volume **(1000+)**, per-board cost drops to under **$1.50**. Custom 2-layer board eliminates **$30+** in redundant components.

PLA Sleeve — FDM 3D Print
CAD-designed and FDM 3D printed in PLA. Print cost: **~$2–4** per unit for materials, with zero tooling cost. Each size variant is a reusable **CAD file** which means no expensive molds are required.

McKibben Actuator — DIY Assembly
Braided mesh tubing **(~$3/m)** over latex tubing **(~$2/m)**. Hand-assembled per unit at prototype scale. At production, the actuator sub-assembly is **$.30-.50** which is a stark price contrast from other treatment options.

Population-Level Economic Impact

COST OF INACTION

Without effective **home rehabilitation**, stroke patients face **extended disability**, higher caregiver **dependency**, increased long-term **care costs**, and **reduced** workforce **participation**. The **economic cost of undertreated stroke disability** vastly exceeds the cost of rehabilitation infrastructure.

Stroke: $36.5B Annual Economic Burden (5)

Including direct **medical costs**, **lost productivity**, informal **caregiving** time, and long-term care **facility costs**. A **10%** reduction via better home rehab = **$3.65B/year** saved.

Long-Term Care Diversion

Skilled nursing facility costs average **$7,908–$9,034/month**. If **PneuFlex** delays or prevents even **1 month** of SNF admission, the $90 device yields a **$7,800+** return per patient.

Hospital Readmission Reduction

Stroke patients with inadequate rehabilitation are **1.5×** more likely to require rehospitalization within **30 days**. Reduced readmissions save hospitals **$15,000–$30,000** per avoided event.

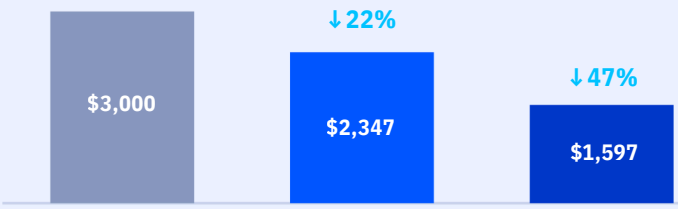
INSURANCE PERSPECTIVE

For Medicare/Medicaid, **PneuFlex** represents an extraordinarily cost-effective intervention **(6)**: a one-time $90 device that displaces recurring **$250/session** clinical costs. From a payer standpoint, **PneuFlex pays for itself** with the *first avoided therapy session.*

COST PROJECTIONS (WITH THERAPY)

SCENARIO	SESSIONS	COST	SAVED
Status quo (no device)	12X	$3000	**$0**
PneuFlex + 9 sessions	9X	$2,347	**$653**
PneuFlex + 6 sessions	6X	$1,597	**$1,403**

REHAB COST PER PATIENT — 12-WEEK PROTOCOL



↓ Up to 47% total episode cost reduction with PneuFlex supplemental home therapy

SNF DIVERSION

RETURN ON INVESTMENT – SINGLE PATIENT

PneuFlex device cost	$90
Avoided clinical sessions (25% reduction over 12-wk)	+$750
1 month SNF diversion **(7)**	+$8,500
Net benefit per patient (conservative)	$9,153

136×
DEVICE COST

$0
COST PER HOME SESSION

AT SCALE: 100,000 PATIENTS

$1.4B
Potential Annual Healthcare Savings

$9M
In Profit (100K units at $90)

5-YEAR MARKET OPPORTUNITY

Addressable US	~500K stroke survivors/yr needing upper-limb rehabilitation
1% Penetration (8)	5,000 devices · $485K revenue at $90 cost → $2–5M at retail price
Global Expansion	15M strokes/yr worldwide · LMICs represent underserved demographics

SYSTEMIC COSTS SAVINGS SUMMARY

COST IMPACT AT SCALE — 100K PATIENTS

Addressable US	$9.7M total
Avoided Clinical Sessions (25%)	$750M saved
SNF Admissions Diverted (est.)	$850M saved
Net Healthcare System Savings	~$1.4B annually

144×
SYSTEM SAVINGS VS. DEVICE COST

$0
AFTER INITIAL PURCHASE

24-7
FUNCTIONALITY

1: The financial benefit gained relative to what was spent
2: When savings or revenue fully offset the initial cost of a product
3: When insurance or government programs repay for covered medical costs
4: Large-scale production that significantly lowers the cost per unit
5: The total financial cost a condition imposes on individuals and society
6: An action that achieves meaningful results at a comparatively low cost
7: Preventing costly nursing facility admissions through earlier, cheaper care
8: The share of a target market a product has successfully reached

Making PneuFlex Unforgettable

VISUAL THEME & PHILOSOPHY

The PneuFlex exhibit uses a **tech aesthetic (1)**: **clean white** surfaces with **rich blue** accents, **OLED-style** data displays, and **circuit-inspired** graphic elements. Every design choice deliberately communicates the **fusion** of medical precision and **embedded engineering** that defines the device. Judges should feel they are looking at a real **medical product**.

COLOR & TYPOGRAPHY (2)

PRIMARY Royal Azure #1866E1	**ACCENT** Capri Blue #00CCFF
SECONDARY Ghost Blue #EBF2FF	**BASE** Deep Ink #040D1F

DISPLAY FONT
-Headers & Statistics
DM Serif Display

BODY FONT
-Labels, Specs, & Data
IBM Plex Mono

PHYSICAL EXHIBIT ELEMENTS

Live Working Prototype
PneuFlex on a foam arm model. Judges can observe McKibben actuation and **hear** the pump inflate.

Live OLED Data Display
Joint angle with **AI confidence** %, **rep counter live** on screen, battery powered, etc.

App Demo Tablet
Battery-powered tablet with live session data syncing over **BLE**.

Trifold backdrop
Blue-white **trifold** with major information and key points about presentation.

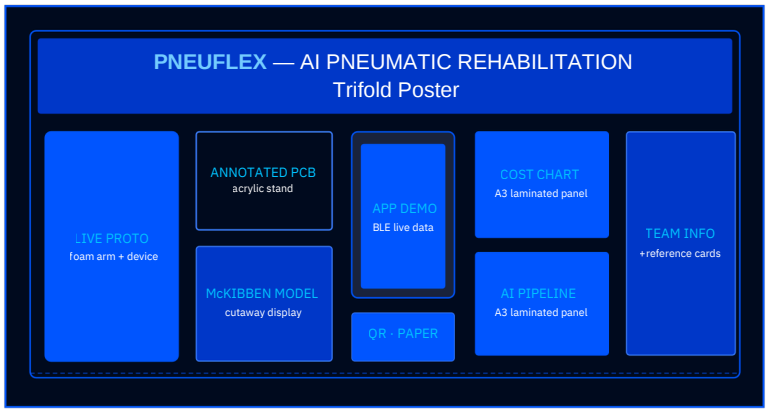
PCB Display Board
Custom **2-layer KiCad PCB** mounted **acrylic stand** + color-coded callout labels for each major component

Printed Data Infographic Panels
A3 laminated panels: cost comparison **chart** and AI inference **pipeline diagram**.

3D Printed PLA Sleeve
CAD-designed, FDM printed in PLA. Evidence of **original engineering design** and not an off-the-shelf component.

Blue LED — Nerve Connection Indicator
Pulses on every detected intent event. Makes the invisible AI decision visible.

TABLE LAYOUT (TOP VIEW NOT TRIFOLD)



ARTISTIC STANDARDS

PRINT	All printed panels on **100lb gloss cardstock**; backdrop on fabric for wrinkle- free presentation; crisp **300dpi** graphics at full scale
LABEL	Every component labeled in **IBM Plex Mono** for a consistent typeface across digital and physical materials creates a **unified brand identity**
CABLE	All cables sleeved in **blue techflex**; connectors heat-shrunk and labeled. **No visible wire clutter** and **professional finish** throughout
MOUNT	PCB displayed on **laser-cut acrylic risers** at a 15° angle. The component side is fully visible, with a **professional product** photography aesthetic

SELF GUIDED VIEWING

Per **HOSA Rule 26**, anyone viewing the exhibit must understand the innovation without a presenter. PneuFlex achieves this through:

(1) the live **OLED** displaying real operational data
(2) **Trifold** explaining all 5 aspects of presentation
(3) labeled **PCB** board
(4) **McKibben** cutaway model showing the **actuation** mechanism
(5) a **QR code** linking to a **90-second** video demo

DIFFERENTIATION
Observers can hear the **pump fire** and see the sleeve contract in real-time. **Proof of concept** no other exhibit can replicate.

TECHNICAL DEPTH
Custom PCB + live **TinyML** inference running throughout presentation. Confidence % is displayed on **OLED**.

MISSION STATEMENT

> **Rehabilitation shouldn't be a privilege. PneuFlex makes clinical-grade recovery accessible to every stroke survivor, everywhere.**

The Story PneuFlex Tells

Why PneuFlex?

Every stroke survivor deserves the same **intensity** of rehabilitation as a clinical patient, regardless of income, geography, or insurance coverage. **PneuFlex** puts that **power** on their arm.

ORIGINALITY STATEMENT

PneuFlex is a wholly **original** design. No commercial or academic device **combines** all three of its core innovations **simultaneously**:

(1) personalized **TinyML intent recognition** trained per-use

(2) **McKibben pneumatic actuation** driven by that AI output

(3) a custom-manufactured **PCB** integrating all control, sensing, power, display, and wireless communication into a single board at **under $100**

COMPETITOR-CREATED ARTWORK

- ✓ All **graphic panels** designed by the team in **Canva** using custom color system and typography.
- ✓ PCB artwork (silkscreen layer **(3)**) designed in **KiCad** which includes **PneuFlex** logo in copper pour.
- ✓ All **system diagrams** and flow charts created by the **team**. No stock infographic templates were used.
- ✓ McKibben cutaway model was physically **constructed**. **Braided mesh + bladder** assembly on the display stand.
- ✓ Device **demo video** (90s) filmed, edited, and captioned, accessible via exhibit **QR code.**

DESIGN PROCESS

Visual Identity

Established the **color system** and **typography** using **Deep Ink (#040D1F)** and Ghost Blue (#EBF2FF) with **DM Serif Display** for headers and **IBM Plex Mono** for data labels to create a unified tech-medical aesthetic that stimulates a medical association with **PneuFlex**.

Content Structure

Organized all panels around a **central problem→solution →proof →impact** discovery. We designed all elements in **Canva** instead of using templates to ensure each section is built clearly and logically. This illustrates the cost and **clinical developments** that **Pneuflex** introduces into society.

Self-Guided Design

Laid out the **trifold** simply to ensure that **PneuFlex** is **understandable** without a presenter. This is done by pairing each panel with a **physical element** on the table: the live OLED, the labeled PCB, and a QR code to the demo video.

WHAT MAKES PNEUFLEX STAND OUT

KINETIC FACTOR

The only exhibit with a live **pneumatic system** that physically moves. Judges can feel and see the McKibben contract. **Tactile** proof of a concept impossible to convey on paper.

AI TRANSPARENCY

The OLED displays **confidence percentages** in real-time. Judges can see the AI making decisions, not just hear claims about it. This proves **PneuFlex's** functionality.

HARDWARE DEPTH

A real **custom PCB**. The annotated board demonstrates professional EE judgment, including: flyback protection, MOSFET selection, TP4056 battery management, and dual-core FreeRTOS.

HUMAN IMPACT CLARITY

$90 displayed prominently against the **$50,000–100,000** exoskeleton benchmark. Immediate understanding of the problem **PneuFlex** solves with no further explanation needed.

EXHIBIT FLOW

Designed to guide judges through a discovery arc **(4)**: problem → solution → proof → impact:

- **Backdrop:** Problem statement + credentials
- **Live Device:** Immediate emotional + intellectual engagement
- **PCB Board:** Engineered from scratch, not assembled from kits
- **Infographic Panels:** Cost + AI pipeline that answer questions
- **App + QR:** Live BLE data + 90s video bridge

CLOSING STATEMENT

PneuFlex is not a concept. It is a **solution**. Every claim in this exhibit can be pointed to on the table. Every number has a source. Every system has a safety layer. This is what the future of **accessible rehabilitation** looks like.

1: A design style mimicking technology products

2: The style and arrangement of text, including font choice

3: Printed text and graphics on a PCB surface used to label components

4: A structure guiding viewers from problem to solution to proof to impact

Literature Cited - APA 7th Edition

REFERENCES

Ang, K. K., Chua, K. S. G., Phua, K. S., Wang, C., Chin, Z. Y., Kuah, C. W. K., Low, W., & Guan, C. (2015). A randomized controlled trial of EEG-based motor imagery brain-computer interface robotic rehabilitation for stroke. *Clinical EEG and Neuroscience, 46*(4), 310–320. https://doi.org/10.1177/1550059414522229

Bütefisch, C., Hummelsheim, H., Denzler, P., & Mauritz, K. H. (1995). Repetitive training of isolated movements improves the outcome of motor rehabilitation of the centrally paretic hand. *Journal of the Neurological Sciences, 130*(1), 59–68. https://doi.org/10.1016/0022-510X(95)00003-B

Centers for Disease Control and Prevention. (2023). *Stroke facts*. U.S. Department of Health & Human Services. https://www.cdc.gov/stroke/data-research/facts-stats/index.html

Edge Impulse. (2024). *Edge Impulse documentation: Getting started with embedded machine learning*. Edge Impulse Inc. https://docs.edgeimpulse.com

Lum, P. S., Burgar, C. G., Shor, P. C., Majmundar, M., & Van der Loos, M. (2002). Robot-assisted movement training compared with conventional therapy techniques for the rehabilitation of upper-limb motor function after stroke. *Archives of Physical Medicine and Rehabilitation, 83*(7), 952–959. https://doi.org/10.1053/apmr.2001.33101

Polygerinos, P., Wang, Z., Galloway, K. C., Wood, R. J., & Walsh, C. J. (2015). Soft robotic glove for combined assistance during task practice and functional motor training. *Journal of NeuroEngineering and Rehabilitation, 12*, Article 48. https://doi.org/10.1186/s12984-015-0046-7

Raj, A., & Shajahan, A. S. (2019). McKibben muscle actuator for soft robotic prosthetics: Design and development. *Materials Today: Proceedings, 18*, 1408–1413. https://doi.org/10.1016/j.matpr.2019.06.641

Stinear, C. M., Smith, M. C., & Byblow, W. D. (2019). Prediction tools for stroke rehabilitation. *Stroke, 50*(11), 3314–3322. https://doi.org/10.1161/STROKEAHA.119.025696

Stroke Association. (2023). *State of the nation: Stroke statistics*. Stroke Association UK. https://www.stroke.org.uk/what-is-stroke/stroke-statistics

Tran, P., Pham, T., Dang, D., & Nguyen, T. (2021). A low-cost wearable pneumatic hand rehabilitation glove. *Sensors, 21*(14), Article 4761. https://doi.org/10.3390/s21144761

Ward, N. S. (2011). Assessment of cortical reorganisation for hand function after stroke. *Journal of Physiology, 589*(23), 5625–5632. https://doi.org/10.1113/jphysiol.2011.214445

Winters, C., van Wegen, E. E. H., Daffertshofer, A., & Kwakkel, G. (2015). Generalizability of the proportional recovery model for the upper extremity after an ischemic stroke. *Neurorehabilitation and Neural Repair, 29*(7), 614–622. https://doi.org/10.1177/1545968314562115

Yap, H. K., Lim, J. H., Nasrallah, F., Goh, J. C. H., & Yeow, C. H. (2015). Soft exoskeleton for hand assistive and rehabilitative therapy using pneumatic actuators. *2015 IEEE International Conference on Rehabilitation Robotics (ICORR)*, 1–6. https://doi.org/10.1109/ICORR.2015.7281200

Johnson & Johnson MedTech. (2024). *The future of rehabilitation technology*. Johnson & Johnson. https://www.jnj.com/innovation/the-future-of-rehabilitation-technology

Cleveland Clinic. (2024). *Stroke rehabilitation: What to expect during your recovery*. Cleveland Clinic Health Library. https://my.clevelandclinic.org/health/treatments/stroke-rehabilitation